Hostel Management System

Software Requirements Document

Team Details

SRN	Name
PES2UG24CS492	Shreyas Patil
PES2UG24CS475	Shobhit Rachit Keshri
PES2UG24CS454	Shalik S
PES2UG24CS491	Shreyas Mohanty

Project Description

A system to manage hostel room allocation and student details.

Tech Stack: Python

Functional Requirements

1. **User Authentication —** Login/logout for admin, warden, and student roles with role-based access control.
2. **Student Registration —** Add, update, and view student personal and academic details.
3. **Room Allocation —** Allocate/reallocate rooms to students based on availability, and track vacant/occupied status.
4. **Room and Block Management —** Add/edit hostel blocks, floors, and room details (capacity, type, fee).
5. **Fee Management —** Record hostel fee payments, generate receipts, and track pending dues.
6. **Complaint Management —** Students can raise maintenance/complaint requests; wardens can view and update status.
7. **Leave/Outpass Management —** Students can apply for leave/outpass; wardens can approve or reject.
8. **Visitor Log —** Record visitor entries against a student's room.
9. **Reports Generation —** Generate reports such as occupancy list, fee due list, and complaint status.
10. **Search and Filter —** Search students/rooms by name, room number, block, or status.

Non-Functional Requirements

1. **Usability —** Simple, intuitive UI usable by non-technical staff (wardens).
2. **Performance —** Room allocation and search queries should respond within 2 seconds under normal load.
3. **Security —** Passwords stored using hashing; role-based access to prevent unauthorized data access.
4. **Reliability —** System should handle concurrent access (multiple students/wardens) without data corruption.
5. **Scalability —** Should support an increasing number of students/rooms without major redesign.

6. **Maintainability —** Modular codebase (MVC pattern) for easy future updates.
7. **Portability —** Should run on common operating systems/browsers without extra configuration.
8. **Availability —** System should be accessible during working hours with minimal downtime.